

Supply Chain Performance & Delay Analytics

Business Problem Statement :

Supply chain operations involve multiple stages, including procurement, inventory management, order fulfillment, shipment and delivery. Delays or inefficiencies at any stage can increase operational costs, reduce customer satisfaction and affect the company's ability to fulfill orders on time.

The company needs to understand where supply chain performance is declining, why deliveries are getting delayed, and which suppliers, warehouses, shipping modes and products require attention.

The main challenge is not only to measure the number of delayed shipments, but to understand the root causes and business impact of delays across the supply chain.

This project analyze supply chain data across orders, suppliers, warehouses, inventory, shipments, delivery events and product categories to identify operational bottlenecks and areas of improvement.

Objectives :

The objective of this project is to evaluate end-to-end supply chain performance, identify delivery and inventory bottlenecks, measure supplier and warehouse reliability, and provide data-driven recommendations to improve OTIF performance, reduce delays, minimize stockouts and improve order fulfillment.

Tools :

Excel was used for data cleaning, validation, KPI calculations and PivotTable-based exploratory analysis. MySQL was used to answer business questions and analyze orders, suppliers, inventory, shipments and delivery performance. Power Query was used for data transformation, while Power BI and DAX were used for data modeling, KPI calculations and development of the interactive supply chain dashboard.

Dataset Overview :

The BharatMart Supply Chain Analytics project uses a relational supply chain dataset designed to analyze the complete flow from customer orders and procurement to inventory, shipment and final delivery.

The dataset combines operational data across customers, products, suppliers, warehouses, orders, purchase orders, inventory, shipments and carriers. This structure allows the analysis to measure order fulfillment, supplier reliability, inventory health, delivery performance and supply-chain delays.

Excel Data Cleaning & Exploratory Analysis

Data Cleaning :

Excel was used as the initial data preparation and exploratory analysis layer of the supply chain project. The raw data was cleaned, validated and transformed before creating KPIs and PivotTables for deeper business analysis.

Data Cleaning Activities :

The following data-cleaning activities were performed to improve data quality and ensure reliable analysis.

- **Removed duplicate** records to avoid double-counting orders, shipments or transactions.
- **Handled missing values** in important fields to maintain data completeness.
- **Standardized** categorical values such as regions, product categories, warehouse names and status fields.
- **Validated data types** for dates, quantities, order values and other numerical fields.
- **Checked invalid or inconsistent values** in operational fields such as delivery status, quantities and performance metrics.
- **Validated relationships** between tables using common keys such as Customer_ID, Product_ID, Order_ID, Supplier_ID, Warehouse_ID and Shipment_ID.
- **Performed final data-quality checks** after cleaning to ensure the dataset was ready for KPI, PivotTable, MySQL and Power BI analysis.

KPI Analysis :

After cleaning the dataset, Excel was used to calculate key supply-chain performance indicators.

- Total Customers
- Total Products
- Total Suppliers
- Total Warehouses
- Total Orders
- Total Order Value
- Total Purchase Orders
- Total Shipments
- Total Quantity Ordered
- Total Quantity Delivered
- Inventory Value
- Overall Supplier On-Time Delivery %
- Stockout Rate %
- Warehouse OTIF %
- Warehouse Utilization %
- Damage Rate %
- Fill Rate %
- Backorder Rate %
- Average Supplier Delay

These KPIs provided a high-level view of delivery performance, supplier reliability, fulfillment efficiency and inventory-related risk.

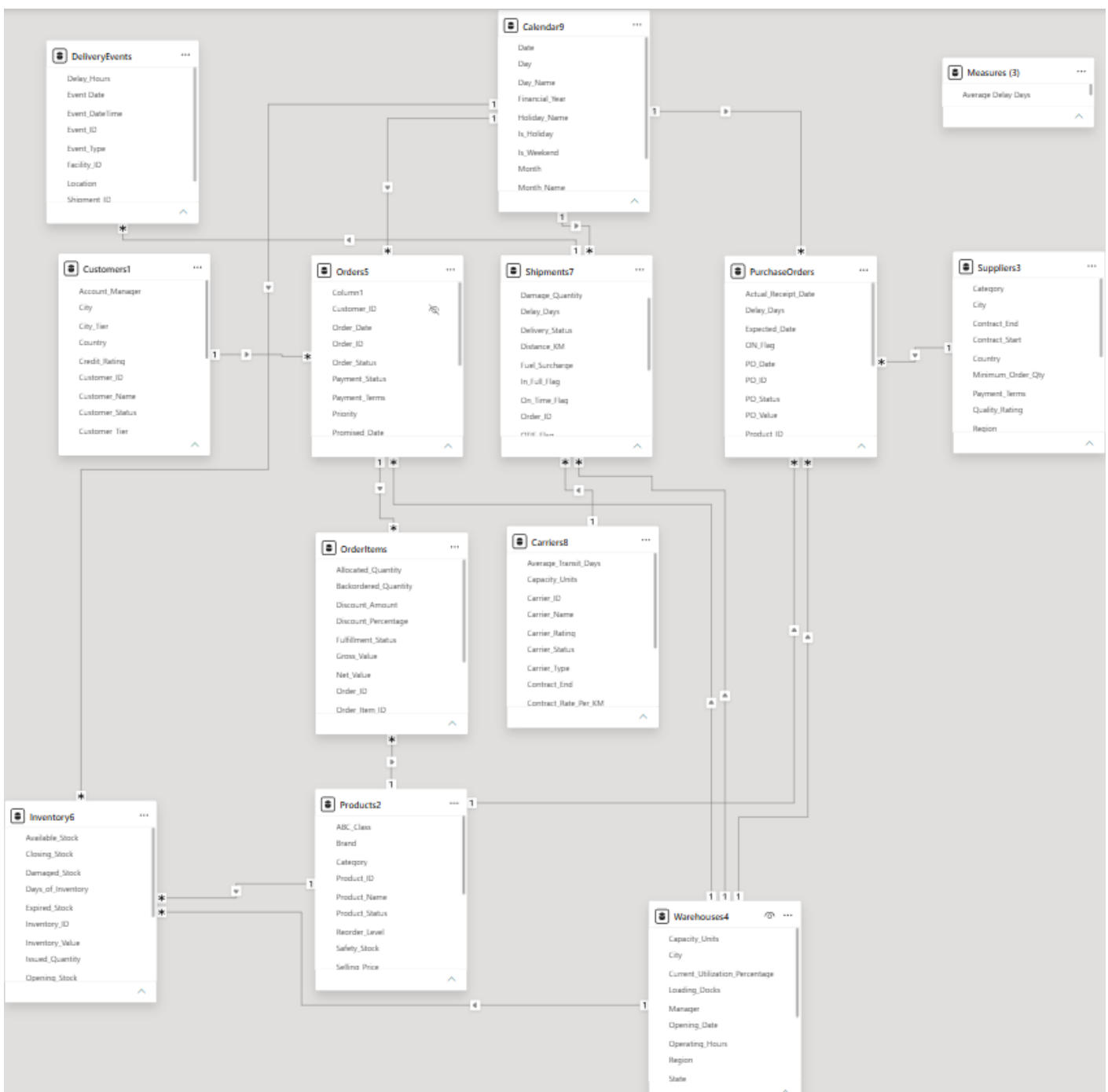
PivotTable Analysis :

PivotTables were created to explore supply-chain performance across different business dimensions and identify operational patterns using relationships.

- **Monthly Order & Sales Performance** : Identifies monthly trends in order volume and sales.
- **Regional Order Performance** : Compares order performance across regions.
- **Regional Supplier Performance** : Evaluates supplier performance across different regions.
- **Warehouse Stockout Analysis** : Identifies warehouses with higher stockout risk.
- **ABC Classification & Inventory Performance** : Analyzes inventory performance based on ABC classification.
- **Delivery Performance by Region & Priority** : Evaluates delivery performance across regions and order priorities.
- **Product Category Fulfillment & Backorder Analysis** : Identifies fulfillment gaps and backorder patterns across product categories.

Power BI Analysis & Dashboard Development

Data Model :



The Power BI data model connects the major supply-chain tables using their respective business keys, allowing metrics from different operational areas to be analyzed together. The model supports analysis across the complete supply-chain flow, from customer orders and procurement to inventory, shipment and final delivery.

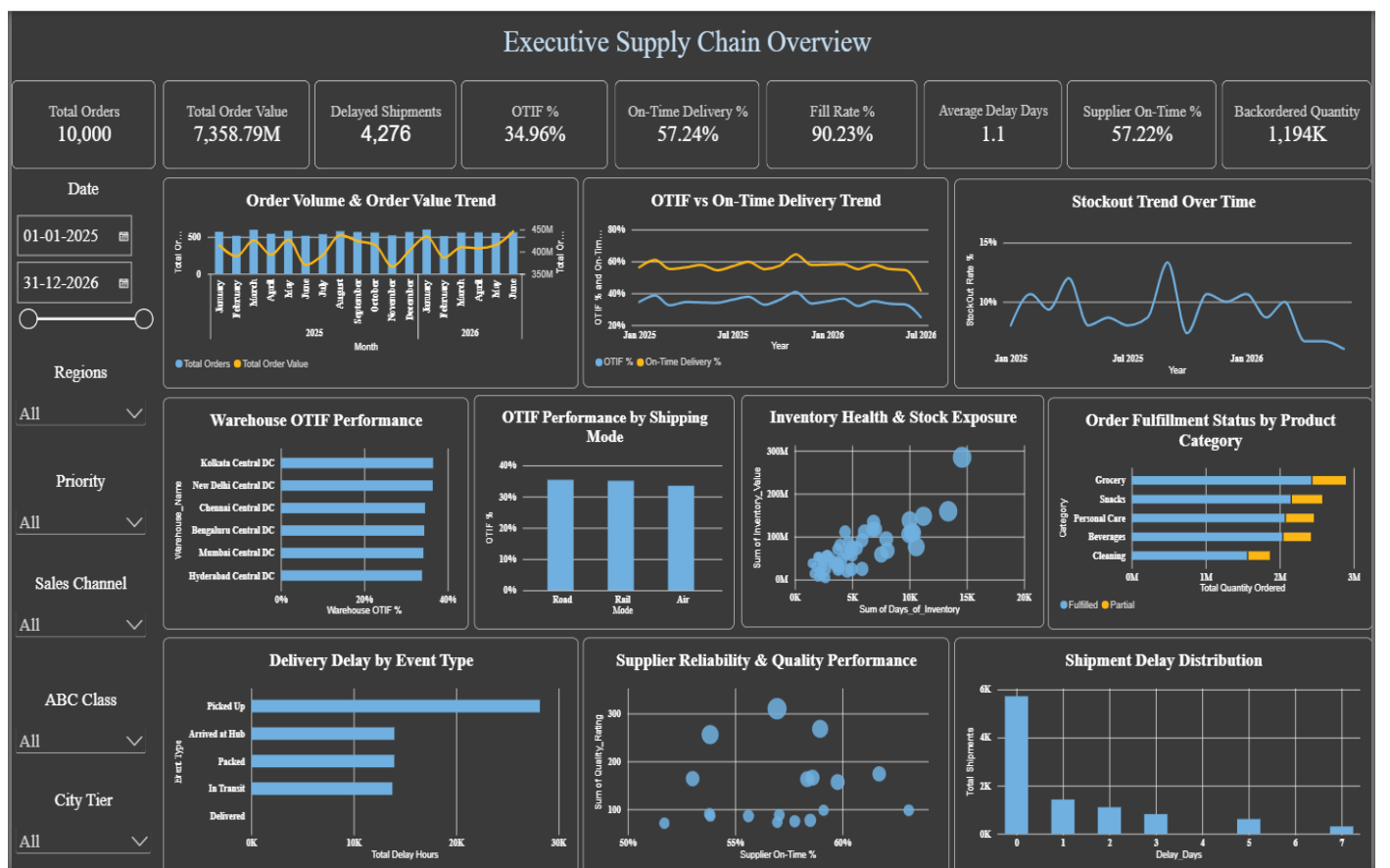
The relationships enable the dashboard to drill into order performance, supplier reliability, warehouse operations, inventory position, fulfillment, backorders and delivery performance while maintaining consistent filtering across the report.

Dashboard KPI'S :

The Power BI dashboard was designed to provide an executive-level view of BharatMart's supply chain performance. KPI cards were created to monitor order volume, financial performance, delivery reliability, fulfillment and inventory-related risks.

- Total Orders
- Total Order Value
- Delayed Shipments
- OTIF %
- On-Time Delivery %
- Fill Rate
- Average Delay Days
- Supplier On-Time %
- Backordered Quantity

Interactive Dashboard :



Power BI was used to transform the cleaned and analyzed supply chain data into an interactive decision-support dashboard. The dashboard brings together order, shipment, warehouse, supplier, inventory and fulfillment metrics to provide a complete view of supply chain performance. Interactive filters allow users to analyze performance across date, region, priority, sales channel, ABC class and city tier. The dashboard helps management identify operational bottlenecks, underperforming warehouses and suppliers, shipment delays, stockout exposure and fulfillment issues, supporting data-driven decisions to improve OTIF performance, delivery reliability, inventory availability and overall supply chain efficiency.

MySQL Analysis & Dataset Limitation

Dataset Limitation :

Due to MySQL Workbench's limitation in handling the large project dataset in my local environment, the original full-sized dataset was not used for executing the SQL analysis. A smaller representative version of the dataset was used to answer and validate the business questions. The SQL queries were written based on the same table structures, relationships and business logic, so the analytical logic can also be applied to the larger dataset when loaded into a suitable database environment.

SQL Analysis Performed on Business Questions :

1. Customer & Segment Analysis
2. Overall Churn Rate Analysis
3. Churn by Customer Segment
4. Revenue Analysis by Subscription Plan
5. Payment Method Analysis
6. Service Usage & Churn Analysis
7. Churn Reason Analysis
8. Customer Satisfaction Analysis
9. Top Customers by Revenue
10. Subscription Plan Revenue Ranking
11. Revenue at Risk from Churn
12. Regional Revenue Risk Analysis

Key Insights :

- 10,000 orders were analyzed with a total order value of approximately ₹735.79M, showing the scale of the supply-chain operations covered by the analysis.
- OTIF performance is 34.96%, indicating that a significant proportion of orders are not being delivered both on time and in full.
- 57.24% of shipments are delivered on time, showing that delivery timeliness is an important area for operational improvement.
- 4,276 delayed shipments were identified, highlighting significant delivery-performance issues across the supply chain.
- The overall fill rate is 90.23%, indicating that most ordered quantities are fulfilled, while the remaining gap represents potential fulfillment issues.
- Supplier on-time performance is 57.22%, indicating that supplier delivery reliability can have a direct impact on downstream operations.

- 1.194M units are backordered, highlighting a significant risk around product availability and order fulfillment.

Key Recommendations :

- Improve OTIF performance by separately monitoring whether shipments are delivered on time and whether the complete ordered quantity is fulfilled.
- Identify and closely monitor underperforming suppliers and establish supplier performance targets based on on-time delivery.
- Analyze the 4,276 delayed shipments to identify recurring delay patterns and operational bottlenecks.
- Improve inventory planning and replenishment for products contributing to high backorder or stockout levels.
- Monitor warehouse-level performance to identify facilities with lower fulfillment and delivery performance.
- Evaluate different shipping modes and carriers to identify more reliable options for time-sensitive orders.

Key Limitations :

- The project uses a sample/simulated supply-chain dataset, so the findings do not represent actual company performance.
- The analysis is based on the available fields and business rules within the dataset; some real-world supply-chain factors may not be captured.
- The analysis identifies patterns and performance gaps, but it does not establish direct causation for every delay or fulfillment issue.
- External factors such as weather, traffic, geopolitical events, supplier disruptions and market conditions are not included in the dataset.
- The project primarily focuses on descriptive and diagnostic analytics rather than predictive supply-chain forecasting.
- The dashboard provides performance visibility, but real-time tracking and automated alerts were not implemented.
- Future analysis could incorporate demand forecasting, delivery-delay prediction and supplier risk scoring to move from monitoring current performance toward proactive decision-making.

Future Scope :

The project can be further enhanced by using Python and machine learning to forecast demand and predict potential delivery delays. Supplier risk scoring can be developed to identify suppliers that are likely to cause future disruptions. Real-time data integration can be added to monitor shipments, inventory and order status continuously. Automated Power BI alerts can help supply-chain teams respond quickly to delays, stockouts and fulfillment issues. These enhancements can transform the project from historical performance analysis into proactive supply-chain decision making.